ModelIt!

Teacher's Guide

# Why Did Dinosaurs Rule and Then Vanish?

Modeling Extinction Dynamics and the Fragility of Dominant Species

## Quick Reference

| | |
|---|---|
| **Grade Level** | Grade 10L2 |
| **NGSS Standard** | HS-LS4-1, HS-LS4-4 |
| **Time** | 45–60 minutes (core lesson) |
| **Materials** | Student Activity Pack, devices with ModelIt! access |
| **Prep** | Print activity packs, test ModelIt! on student devices |
| **Grouping** | Partners or small groups (2–4) |


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# 1 NGSS Standards & CAST Alignment

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**Performance Expectation: HS-LS4-1, HS-LS4-4**

Communicate scientific information that common ancestry and biological evolution are supported by multiple lines of empirical evidence. Construct an explanation based on evidence for how natural selection leads to adaptation of populations.

## Three-Dimensional Learning

| | |
|---|---|
| **Science Practice** | **Constructing Explanations** — Students construct evidence-based explanations of mass extinction dynamics by analyzing the interactions between environmental change rate, adaptation speed, and survival traits in their computational model. |
| **Disciplinary Core Idea** | **LS4.A Evidence of Common Ancestry / LS4.C Adaptation** — Multiple lines of fossil evidence document mass extinction events and subsequent adaptive radiation; natural selection drives adaptation but cannot operate faster than the rate of environmental change. |
| **Crosscutting Concept** | **Cause and Effect** — Students trace the causal chain from catastrophic environmental change through food web collapse, competitive hierarchy reversal, and selective survival to post-extinction recovery and adaptive radiation. |

## CAST Testing Alignment

**MULTIPLE CHOICE:**

A mass extinction event rapidly eliminates 75% of species. Based on your model, which type of organism is most likely to survive?

**CONSTRUCTED RESPONSE:**

Using your model, explain why dinosaurs — the most successful land animals in Earth's history — went extinct while small mammals survived. Reference Environmental Change Rate, Adaptation Speed, and Survival Advantage in your answer.

# 2 Scoring Rubric

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### Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

### Scoring Guidance

**Expert (4):** Student demonstrates deep understanding and can transfer knowledge to new contexts.
**Proficient (3):** Student meets grade-level expectations and shows solid understanding.
**Developing (2):** Student is making progress but needs additional scaffolding or practice.
**Beginning (1):** Student needs significant support; consider using Differentiation strategies (Section 7).

# 3 Background Content Knowledge

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**For Your Confidence**

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Dinosaurs were not failures. They were the most successful terrestrial vertebrates in Earth's history, dominating every continent for 165 million years — 800 times longer than modern humans have existed. Their extinction was not a result of evolutionary inferiority but of environmental change so sudden and severe that adaptation was physically impossible.

## The Chicxulub Impact

Sixty-six million years ago, a 10-kilometer asteroid struck what is now the Yucatan Peninsula at 70,000 km/h. The impact released energy equivalent to 10 billion nuclear weapons. Within hours, the entire hemisphere was subjected to superheated debris raining from the sky, igniting global wildfires. Within weeks, sulfur aerosols and dust blocked 90% of sunlight. Within months, photosynthesis largely ceased, collapsing food webs from the bottom up. Global temperatures dropped 10-15 degrees Celsius. This 'impact winter' lasted 3-10 years.

## Why Size Killed the Dinosaurs

Large body size was the dinosaurs' greatest advantage — and their fatal vulnerability. Large animals require enormous caloric intake. A Tyrannosaurus rex needed an estimated 40,000-60,000 calories per day. When photosynthesis collapsed and food webs disintegrated, large animals simply could not find enough food to sustain their metabolism. Small mammals could survive on insects, seeds, and detritus. Small body size also meant shorter generation times, lower caloric needs, ability to burrow for shelter, and capacity to enter torpor (a hibernation-like state) during resource scarcity.

## The Survivors

The organisms that survived the K-Pg extinction share revealing traits. Mammals (small, burrowing, insectivorous), birds (small-bodied descendants of theropod dinosaurs), crocodilians (aquatic, able to survive long periods without food), turtles (protected by shells, slow metabolism), frogs (small, aquatic, can estivate), and insects (short generation times, diverse diets, small body size). The pattern is clear: generalist diet, small body size, low metabolic demands, ability to shelter or go dormant, and short generation times were the winning combination.

## Recovery and Adaptive Radiation

After the extinction pressure lifted, the surviving lineages underwent explosive adaptive radiation — rapid evolutionary diversification into empty ecological niches. Mammals evolved from small, mouse-like creatures into every modern form: whales, bats, elephants, primates. Birds diversified from small, ground-dwelling survivors into 10,000+ species occupying every habitat from ocean to mountaintop. This recovery took millions of years — the first large mammals didn't appear until 10 million years after the impact. Evolution rebuilds, but never quickly.

# 4 The LEVER Framework

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### About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

**L** **Locate the System**

Students identify seven components of the extinction dynamics system: Species Diversity, Environmental Change Rate, Adaptation Speed, Extinction Pressure, Niche Competition, Survival Advantage, and Population Recovery — with Environmental Change Rate as the external driver.

**E** **Establish Relationships**

Students determine that Environmental Change Rate drives Extinction Pressure, which overwhelms Adaptation Speed when change is too rapid. Niche Competition intensifies as resources shrink. Survival Advantage shifts from traits favoring dominance to traits favoring resilience. Population Recovery begins when conditions stabilize.

**V** **Visualize & Model**

Students build a computational model showing how the rate of environmental change relative to adaptation speed determines whether species survive or go extinct.

**E** **Evaluate Outcomes**

Students run gradual change, asteroid impact, and post-extinction recovery scenarios to compare outcomes and identify which biological traits provide survival advantage in each context.

**R** **Revise & Extend**

Students add Food Web Complexity, Genetic Diversity, or Geographic Range to explore additional factors that determine species vulnerability and resilience during mass extinction events.

## Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

| | |
|---|---|
| **At the start of a unit** | Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it. |
| **After a reading or lab** | Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes. |
| **As a “what-if” tool** | When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back. |
| **For peer discussion** | Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence. |
| **As a presentation tool** | Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time. |
| **At the end of a unit** | Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved. |

# 5 Slide-by-Slide Facilitation

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## Slide 1: Cover

2 min

**Say:** *"For 165 million years, dinosaurs ruled every continent on Earth. Then, in a single bad day, it was over. What determines who lives and who dies when the world changes overnight?"*

**Watch For**

Show the scale: dinosaurs ruled for 165 million years; modern humans have existed for 200,000. Ask: What can a 165-million-year winning streak not protect you from?

## Slide 2: Learning Objectives

3 min

**Say:** *"Today you're modeling the most dramatic event in the history of life — and asking a question that matters for the future: what determines survival?"*

**Watch For**

Have students read objectives. Pre-teach 'mass extinction' and 'adaptive radiation.' Quick-write: List three traits you think would help a species survive a catastrophe.

## Slide 3: Big Question

3 min

**Say:** *"Being big, powerful, and dominant for 165 million years wasn't enough. What went wrong — and what does it teach us about survival?"*

**Watch For**

Think-pair-share: Students brainstorm why being the top predator might become a disadvantage. Share and discuss.

## Slide 4: LEVER Framework

2 min

**Say:** *"We're going to model what happens when the environment changes faster than evolution can respond. Spoiler: it's not pretty."*

**Watch For**

Preview each LEVER step. Emphasize the critical variable: the ratio of environmental change rate to adaptation speed.

## Slide 5: Activity 1: Components

8 min

**Say:** *"Seven factors determine whether a species makes it through a mass extinction. Only one is external — the thing that changed. Everything else is how life responds."*

**Watch For**

Guide component sorting. Discuss why Environmental Change Rate is external (the asteroid) while all biological responses are internal.

## Slide 6: Activity 2: Connections

8 min

**Say:** *"When the asteroid hits and food webs collapse, trace what happens to every component. Where does the chain of death begin — and where does survival emerge?"*

**Watch For**

Help students map the cascade from environmental change through extinction pressure, niche competition collapse, and survival advantage reversal.

## Slide 7: Activity 3: Simulation

10 min

**Say:** *"Three scenarios: slow change, catastrophic change, and the aftermath. Watch how completely different the survival rules become when the change rate shifts."*

**Watch For**

Students predict outcomes first. Run all three scenarios. Compare which traits win in gradual vs. catastrophic change. Discuss the post-extinction recovery pattern.

## Slide 8: Discoveries

5 min

**Say:** *"The biggest, strongest, most dominant animals died. The small, flexible, generalist survivors inherited the Earth. What does that tell us about our own species?"*

**Watch For**

Lead discussion connecting extinction dynamics to modern biodiversity crisis. Are humans acting as the asteroid for current species?

## Slide 9: STEM Challenge

12 min

**Say:** *"Conservation biologists need to know which species are most at risk RIGHT NOW. Design the scoring system that identifies them before it's too late."*

**Watch For**

Groups design resilience indices using model insights. Score real modern species and predict which are most vulnerable. Present and defend rankings.

# 6 Answer Key & Expected Responses

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## Component Sorting

- **EXTERNAL:** Environmental Change Rate
- **INTERNAL:** Species Diversity, Adaptation Speed, Extinction Pressure, Niche Competition, Survival Advantage, Population Recovery

## Relationships

- **Environmental Change Rate to Extinction Pressure:** POSITIVE (+) — Faster Environmental Change Rate increases Extinction Pressure because organisms have less time to adapt, migrate, or find alternative resources. When change outpaces the biological capacity to respond, populations decline toward extinction.
- **Extinction Pressure to Species Diversity:** NEGATIVE (-) — Higher Extinction Pressure directly reduces Species Diversity as populations that cannot survive the changed conditions decline and disappear. The most specialized species — those tightly adapted to pre-change conditions — are eliminated first.
- **Species Diversity to Population Recovery:** POSITIVE (+) — Greater remaining Species Diversity after the extinction event provides more genetic and ecological raw material for Population Recovery. Surviving lineages with diverse traits and wide ecological tolerances can radiate into empty niches more rapidly.

## Key Discoveries

- Mass extinction is not about which species are the most powerful but which are best suited to survive catastrophic conditions — size and dominance become liabilities when food webs collapse
- Adaptation Speed has a biological limit determined by generation time and genetic variation — when Environmental Change Rate exceeds this limit, extinction is inevitable regardless of fitness
- Survival Advantage shifts dramatically during mass extinction events — traits like small body size, generalist diet, and ability to enter torpor become critical, reversing the normal competitive hierarchy
- Population Recovery after mass extinction follows a predictable pattern: opportunistic generalists colonize first, followed by gradual specialization and adaptive radiation over millions of years

# 7 Collaborative Learning Protocols

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### Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

### Model Gallery Walk

**When:** After students build models (LEVER Phase V)
**Setup:** Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
**Protocol:** Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
**Why it works:** Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

### Scientist Fishbowl

**When:** During simulation discussion (LEVER Phase E)
**Setup:** 4-5 students sit in center circle. Rest of class observes with clipboards.
**Protocol:** Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
**Why it works:** Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

### Component Jigsaw

**When:** During component analysis (LEVER Phase L)
**Setup:** Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
**Protocol:** Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
**Why it works:** Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

### Prediction Debate

**When:** Before running simulations (LEVER Phase E)
**Setup:** Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
**Protocol:** Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
**Why it works:** Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

### Pair Programming for Models (Digital Collaboration)

**When:** Throughout model building
**Setup:** Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
**Protocol:** Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
**Why it works:** Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

### Quick Protocol Selection Guide

| | |
|---|---|
| **Short on time (5 min)** | Prediction Debate — gets bodies moving, generates energy fast |
| **Full period available** | Component Jigsaw — deep collaboration, every student contributes |

| | |
|---|---|
| **After simulation day** | Model Gallery Walk — peer feedback, multiple perspectives |
| **Advanced discussion** | Scientist Fishbowl — builds academic discourse skills |
| **Every day default** | Pair Programming — transforms individual work into dialogue |

# 8 Differentiation & Extensions

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### Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

### Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

### English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

### Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

# 9 Common Misconceptions

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### Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

| STUDENTS MAY THINK | SCIENTIFIC UNDERSTANDING |
|---|---|
| "Dinosaurs went extinct because they were inferior" | Dinosaurs were extraordinarily successful organisms that dominated Earth for 165 million years — far longer than mammals have been dominant. They didn't go extinct because of evolutionary inferiority but because a 10-kilometer asteroid changed the environment faster than any organism could adapt. The same event would extinguish most modern species if it happened today. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "Evolution can save a species from any threat" | Natural selection can only operate as fast as organisms can reproduce and transmit genetic changes to offspring. When environmental change occurs in months (asteroid impact) but adaptation requires thousands to millions of generations, evolution is powerless. There is a fundamental speed limit to adaptive evolution set by generation time, population size, and the magnitude of change required. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "All dinosaurs went extinct" | Birds ARE dinosaurs — specifically, they are the surviving lineage of theropod dinosaurs. Modern birds descended from small, feathered dinosaurs that survived the K-Pg extinction. With over 10,000 species, dinosaurs are actually among the most diverse vertebrate groups alive today. When you see a pigeon, you are seeing a dinosaur. |